Adiabatic gate teleportation

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This talk: PRL 2009
Related work by Oreshkov, PRL 2009, poster #13, tonight
Applications to Cluster States: arXiv:0912.2098

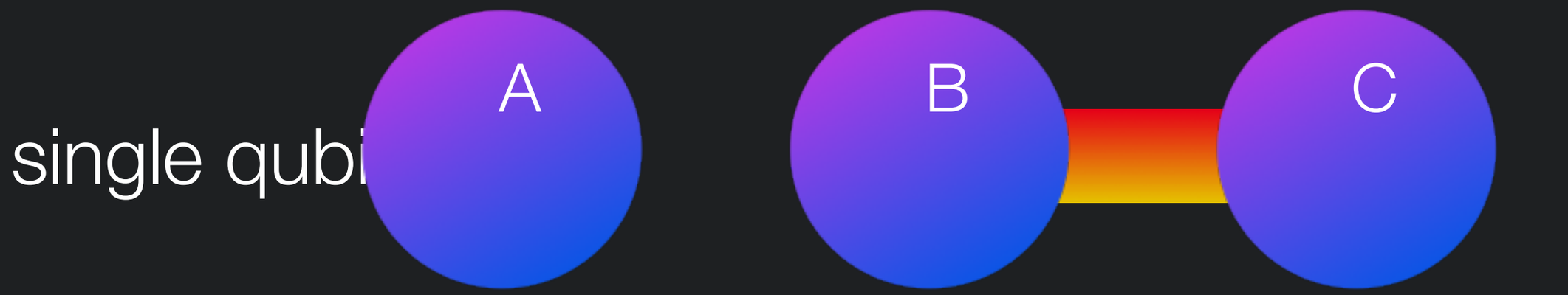
Adiabatic gate teleportation

Dave E, Melissa A and M





Adiabatic teleportation



$|\psi\rangle$ $|\Phi\rangle$

This is a ground state of $H_i = -X_2X_3 - Z_2Z_3$

(could also use the exchange interaction)

Adiabatic teleportation



This is a ground state of $H_f = -X_1X_2 - Z_1Z_2$

Adiabatic teleportation



$|\psi\rangle$ $|\psi\rangle$

$$H(t) = (1-t)H_i + tH_f$$

The adiabatic evolution acts like a post-selected teleportation!

$$\mathcal{T}\exp\left(-i\int_0^t \mathrm{d}\tau H(\tau)\right)$$

Notice that the ground space is stabilized by XXX and ZZZ for all t.

Adiabatic *gate* teleportation



$$|\psi\rangle \qquad\qquad U_3|\psi\rangle$$

$$U_3 H(t) U_3^\dagger = (1-t) U_3 H_i U_3^\dagger + t H_f$$

Now the adiabatic evolution ***teleports*** the unitary onto the qubit.

Gottesman Chuang 1999

Universality

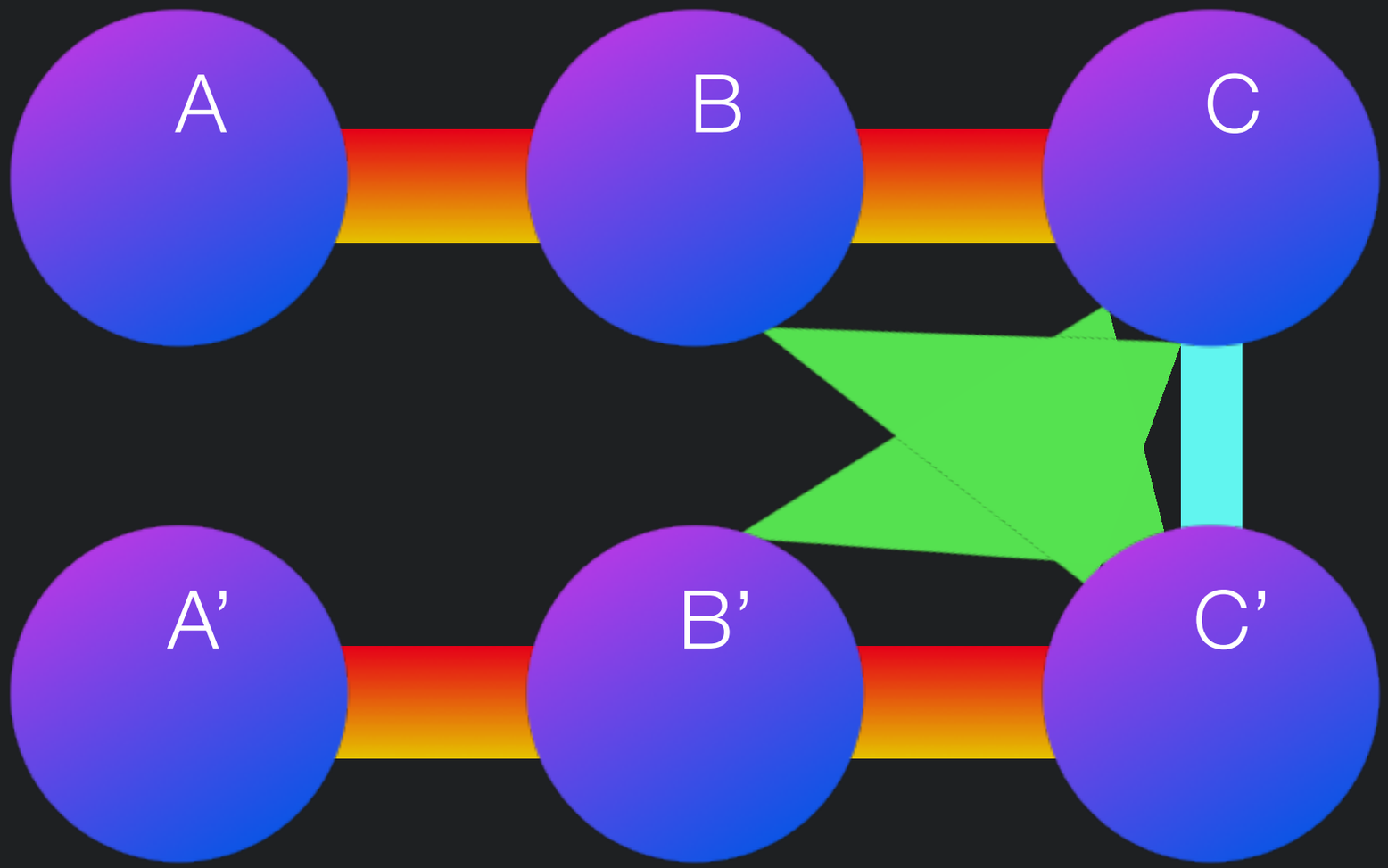


$U_b U_a |\psi\rangle$ $U_a |\psi\rangle$

Etc...

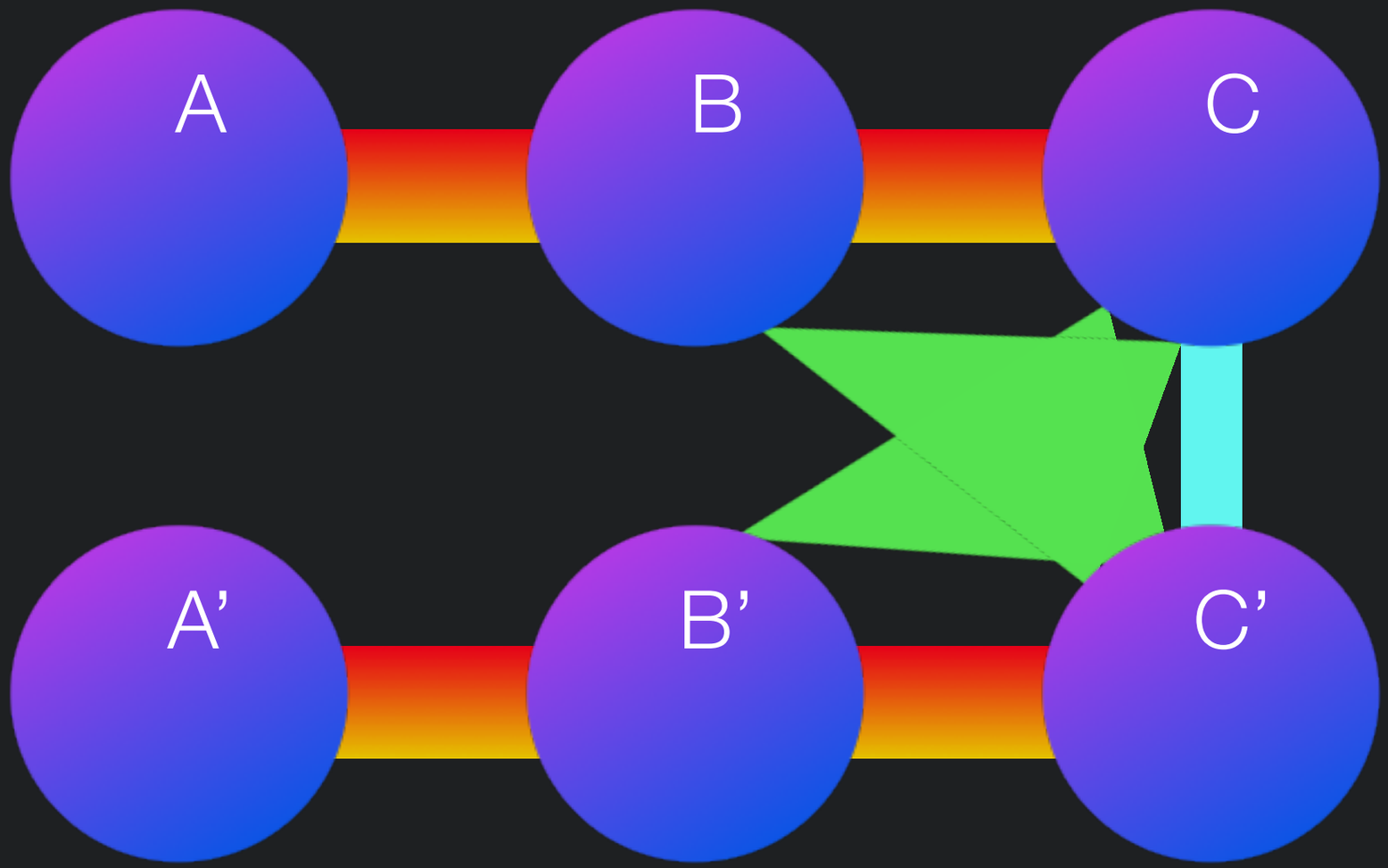
but what about two qubit gates?

Universality



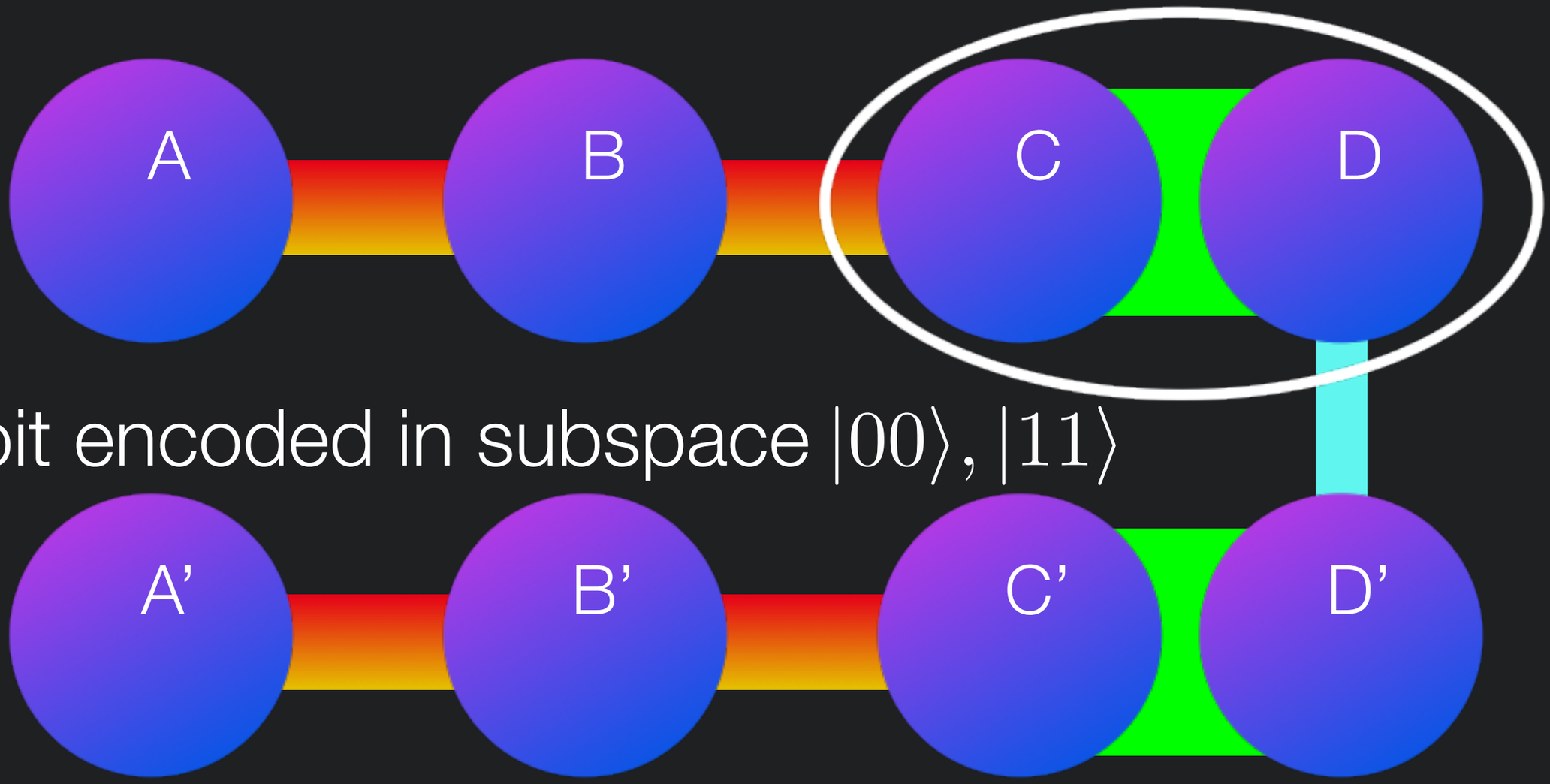
Two-qubit gates introduce 3-body terms...

Universality



Two-qubit gates introduce 3-body terms...
to get rid of them, use perturbation gadgets.

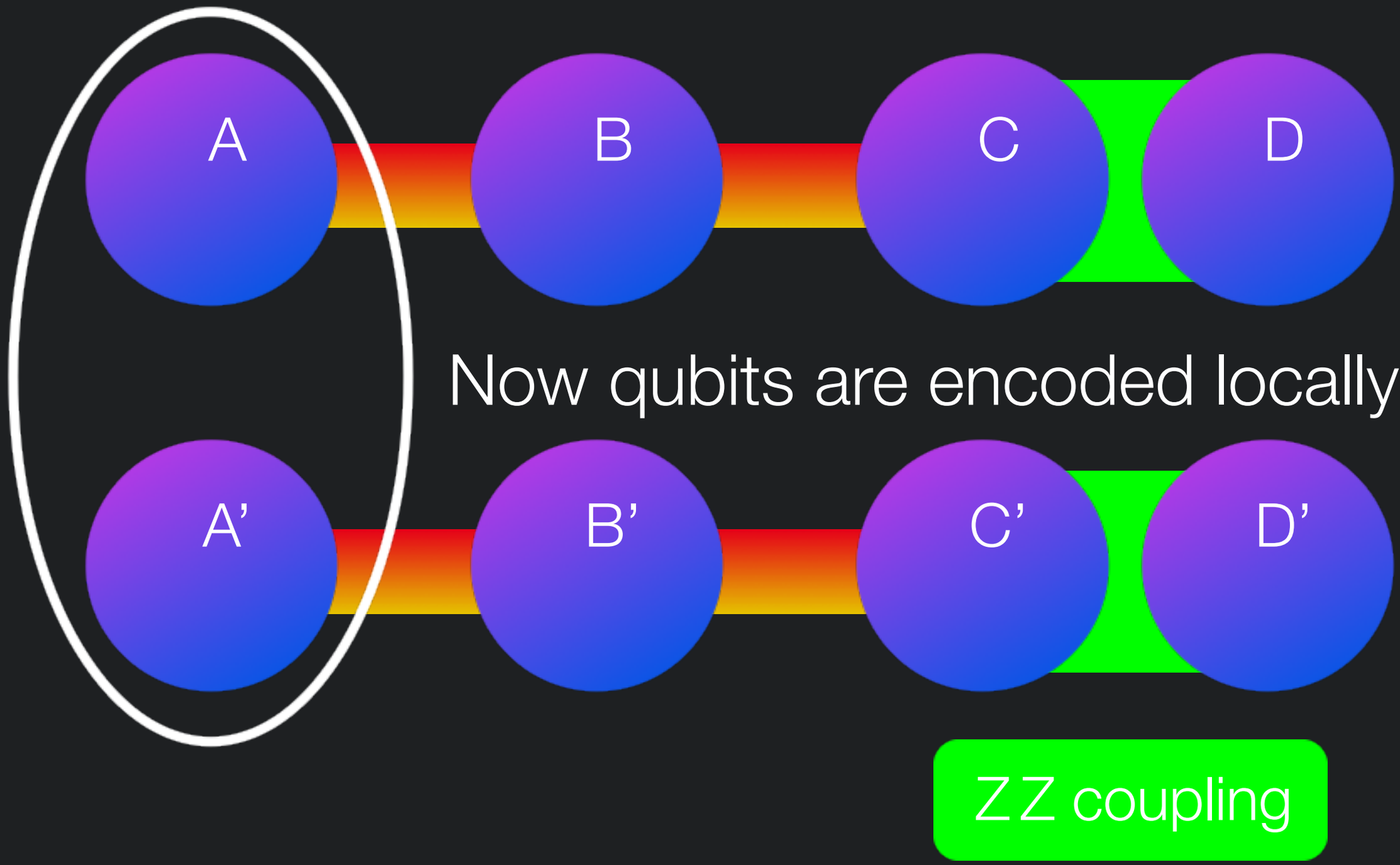
Universal, 2-body



Qubit encoded in subspace $|00\rangle, |11\rangle$

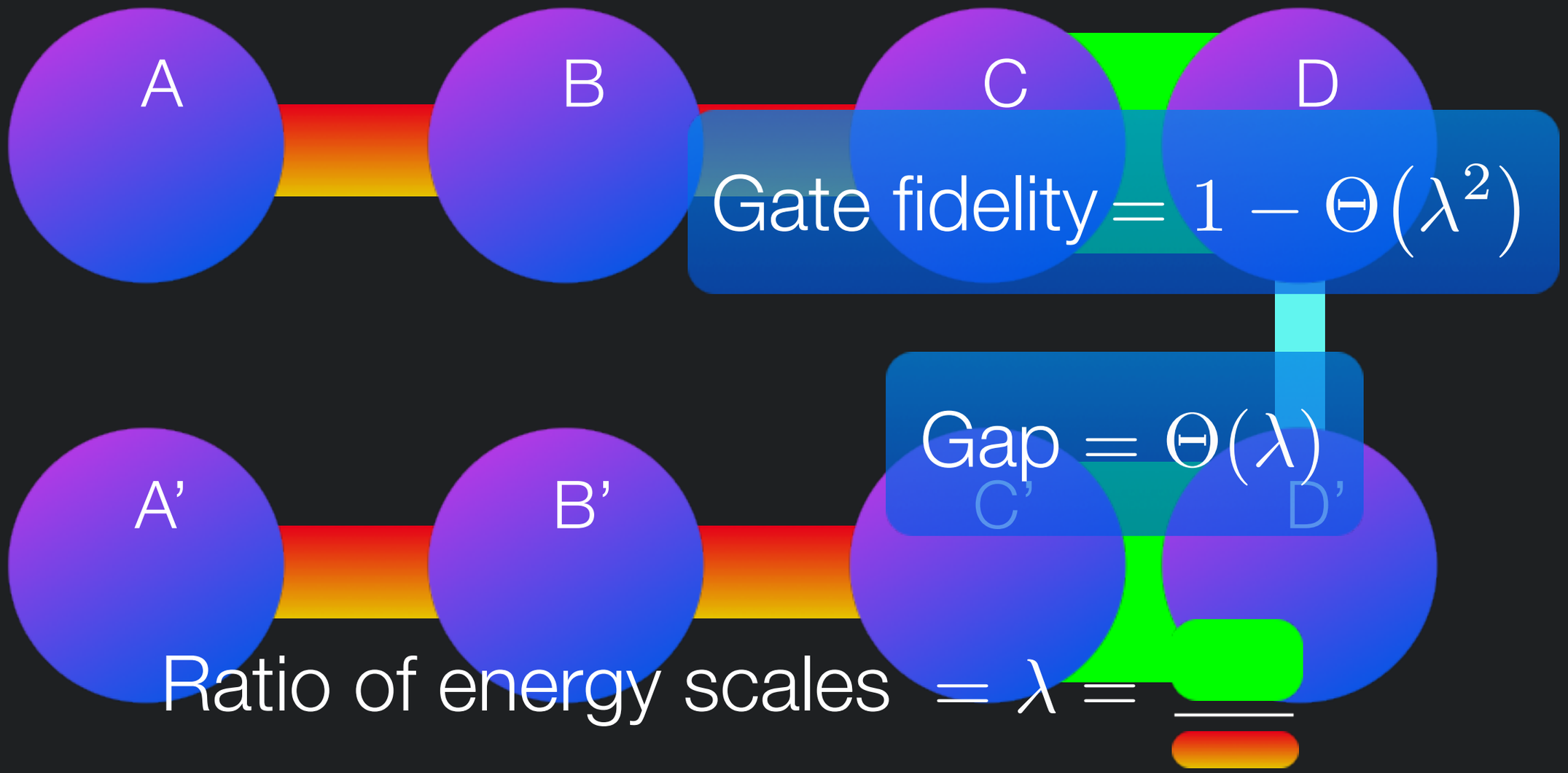
Our perturbation gadgets: Bartlett & Rudolph 2006

Universal, 2-body



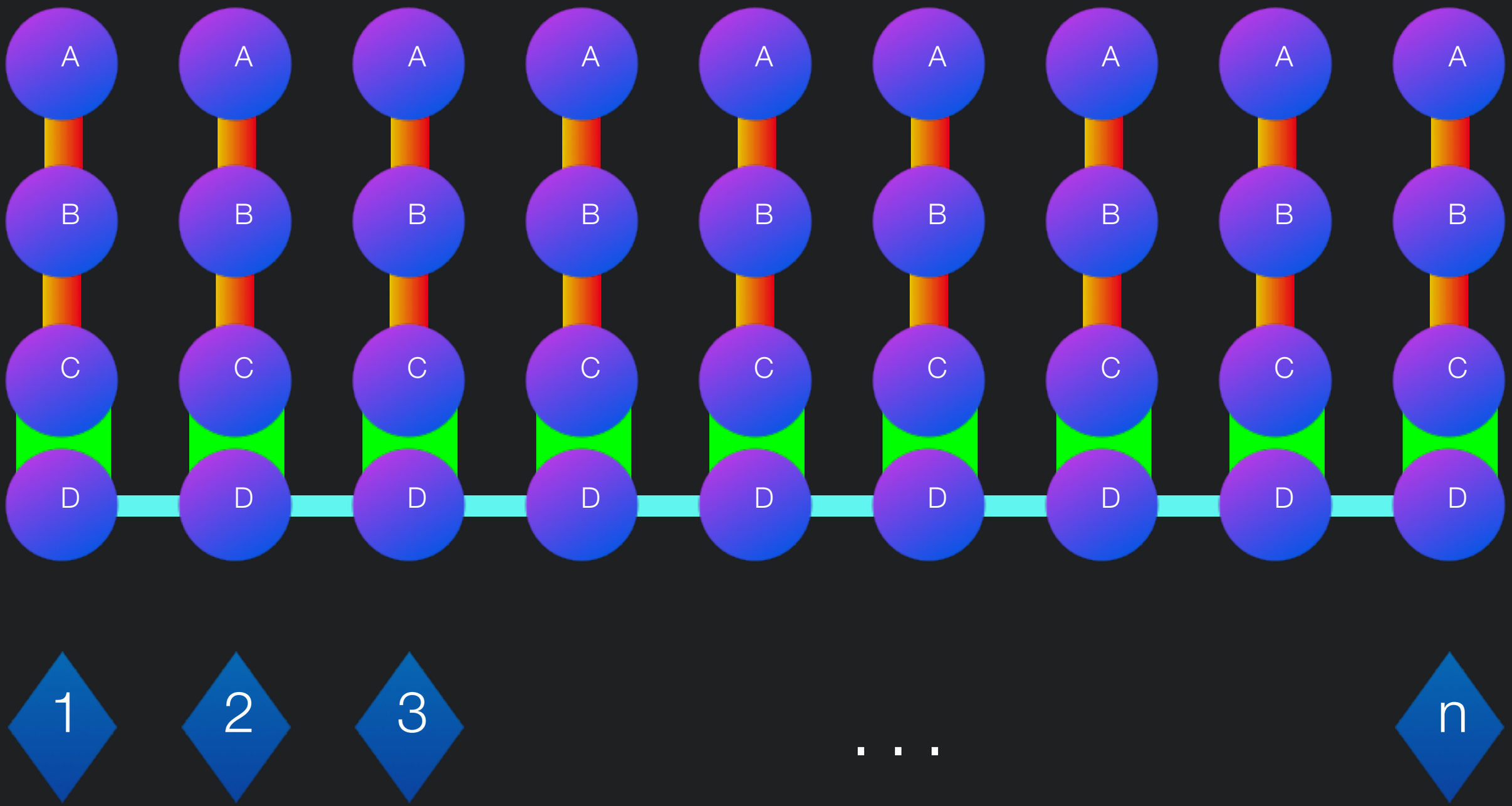
Our perturbation gadgets: Bartlett & Rudolph 2006

Universal, 2-body



Our perturbation gadgets: Bartlett & Rudolph 2006

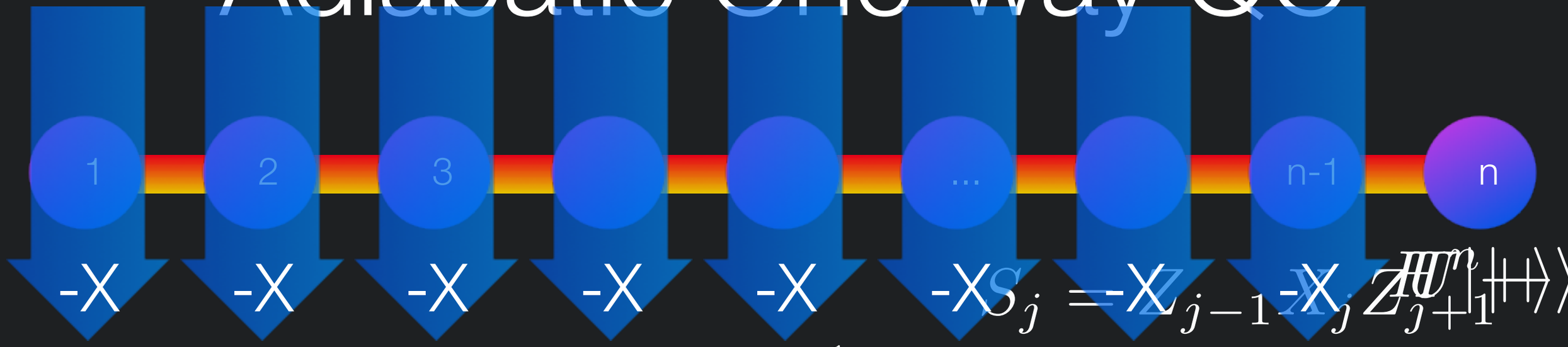
1-d architecture



Why is this interesting?

- Adiabatic or holonomic evolution offers robustness to timing and control errors that exist in the circuit model
- Excitations are suppressed by the constant gap
- “Ground state” errors can be corrected via coding
- It is modular, and hence as easy to program as the circuit model
- Uses only control between subsystems, not levels
- Gates are prepared offline, leading to fewer errors
- It leads to more results of interest to theorists...

Adiabatic One-way QC



$$S_j = -X_{j-1} Z_j X_{j+1}$$

$$|\Psi\rangle$$

$$n-1$$

$$H = -Z_{n-1}X_n - \sum_{j=2} S_j$$

Suppose we prepare $|+\rangle$ on the first physical qubit

Turn on -X fields and turn off cluster state coupling

Rotating the X fields in X-Y plane to make it universal

The gap is still constant

Quantum Transistors



$$H(t) = (1-t)H_C + tH_X$$

This is the transverse-field Ising model (with funny BCs)

The gap is $= \Theta(1/n)$

Reminiscent of transistors:
an applied field induces a quantum phase transition
that yields a quantum logic gate!

Summary & Open Problems

- Adiabatic evolution is robust to timing and control errors and excitations to excited states
- AGT lets us prepare gates offline and perform dynamics adiabatically

- Explicitly gadgetize the one-way version
- Other ground-space computing schemes?
- Connections with topological QC
- Fault tolerance: Oreshkov Brun Lidar 2009, Oreshkov 2009

Holonomic QC: Zanardi Rasetti 1999, Pachos Zanardi Rasetti 1999